AI for Compassion: How Large Language Models Could Support Moral Circle Expansion and Suffering Reduction

A Suffering-Focused Perspective on Moral Circle Expansion and AI-Assisted Moral Reflection.

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August 2026 | Version 1.0

Conceptual research paper. This manuscript has not undergone peer review.

Abstract

Large language models (LLMs) are increasingly used to explain complex ideas, support reflection, and help users navigate difficult decisions. This paper asks whether those capabilities could be used to increase moral concern for beings whose suffering is routinely neglected. The question is approached from a suffering-focused ethical perspective while remaining open to disagreement about the underlying moral theory. We review the idea of moral circle expansion and several psychological mechanisms that can limit concern for distant, numerous, or non-human sufferers, including scope insensitivity, psychic numbing, identifiable-victim effects, and speciesism. We then examine four possible mechanisms through which conversational AI could support moral reflection: Socratic questioning, evidence-grounded perspective-taking, numerical and contextual reframing, and practical action support. On this basis, we propose a conceptual system called CompassionGPT, designed to encourage reflection without coercion or shame. We also identify important failure modes, including factual hallucination, persuasive overreach, reinforcement of existing biases, anthropomorphism, attitude-behavior gaps, and the possibility that attempts to promote compassion could themselves create harms. Rather than claiming that LLMs can make people more compassionate, we argue for a narrower and testable proposition: appropriately designed conversational systems may be useful tools for moral reflection if their claims are evidence-grounded, their persuasive power is constrained, and their effects are evaluated through behavioral as well as attitudinal measures. The paper concludes with a research agenda for controlled studies of moral circle expansion and suffering-reduction interventions using AI.

Keywords: large language models; moral circle expansion; suffering-focused ethics; animal welfare; moral psychology; AI ethics; moral persuasion; sentience; compassion; human-AI interaction

1. Introduction

Large language models are becoming general-purpose interfaces for information, reflection, and decision support. Most discussion of their social value has focused on productivity, education, health information, or economic transformation. A less explored possibility is whether conversational AI could help people notice forms of suffering that ordinary attention tends to overlook.

This possibility matters from a suffering-focused ethical perspective. Suffering-focused ethics (SFE) is a family of moral views that gives especially strong weight to preventing and alleviating suffering. These views differ about whether positive welfare has independent value, how suffering should be compared across individuals, and what trade-offs are permissible. The common thread relevant to this paper is the claim that severe suffering deserves substantial moral attention and should not be neglected merely because its victims are distant, numerous, politically weak, or non-human (Vinding, 2020).

The paper does not attempt to establish SFE as the correct moral theory. Instead, it asks a more limited practical question: if a person already accepts that severe suffering deserves serious consideration, or is open to that idea, can an LLM help them reason more consistently about whose suffering counts and what can be done about it?

This question connects two literatures that are rarely considered together. The first concerns moral circle expansion: the psychological and cultural process through which moral concern is extended to people or beings previously treated as outside the community of moral consideration (Singer, 1981; Crimston et al., 2016). The second concerns conversational AI and its ability to tailor explanations, ask questions, generate examples, and adapt to a user's responses. The central proposal is that these capabilities could be combined into a tool for structured moral reflection.

The claim is deliberately modest. An LLM does not automatically produce wisdom, compassion, or reliable moral judgment. It can also persuade people toward false conclusions, reproduce social prejudices, and present uncertain claims with unwarranted confidence. The useful research question is therefore not whether AI is inherently compassionate, but under what design and governance conditions it could support human concern for suffering without becoming a manipulative moral authority.

Scope and contribution: This paper is a conceptual, literature-informed design study rather than an empirical effectiveness study. It asks whether large language models could support moral reflection and moral-circle expansion while avoiding coercion and serious unintended harms. The paper develops this possibility as a testable hypothesis and proposes mechanisms, a prototype architecture (CompassionGPT), safeguards, and an empirical evaluation agenda. No original participant study is reported.

2. Moral Circle Expansion and Suffering

The term “moral circle” refers to the set of beings whose interests a person regards as morally relevant. The boundaries of that circle vary across individuals and cultures, and they can change over time. Historical examples include the expansion of legal and moral concern across social classes, ethnic groups, genders, and other previously marginalized groups (Singer, 1981; Nussbaum, 2006).

The idea should not be interpreted as a simple linear story of inevitable moral progress. Human history contains both expansions and contractions of concern. Nevertheless, the concept is useful because it highlights a recurring ethical question: why should morally relevant interests stop at a particular boundary?

For suffering-focused ethics, sentience is especially important because suffering is an experiential phenomenon. From a suffering-focused perspective, if a being can plausibly experience pain, fear, distress, or other negatively valence states, excluding that being from moral consideration requires additional justification. This does not by itself settle every question about comparative moral status, but it challenges species membership as a sufficient reason for ignoring suffering.

This issue is particularly important for non-human animals. Humans routinely distinguish between companion animals and farmed animals even when there are relevant similarities in their capacity for pain, fear, learning, and social behavior. Psychological research on speciesism and the “meat paradox” suggests that people can simultaneously endorse animal welfare and participate in practices that impose substantial harms on animals (Bastian & Loughnan, 2017; Caviola et al., 2019).

A wider moral circle also raises difficult questions about wild animals and possible future artificial minds. These areas involve substantial uncertainty. In the case of wild animals, interventions can themselves disrupt ecosystems or create unintended harms. In the case of AI, there is currently no established scientific basis for treating ordinary LLMs as conscious sufferers. The appropriate stance is therefore precautionary rather than anthropomorphic: investigate the possibility of artificial sentience without assuming that present-day chatbots have subjective experiences (Birch, 2024; Metzinger, 2021).

The practical significance of moral circle expansion is that moral attention is a scarce resource. People cannot respond to every instance of suffering personally. A useful AI system could therefore have a role in helping users identify neglected interests, compare competing claims, and move from abstract concern toward proportionate action.

3. Why Suffering Is Easy to Neglect

A central obstacle is not necessarily a lack of sympathy but the way human attention and judgment respond to scale, distance, and abstraction.

Scope insensitivity and psychic numbing. People often struggle to increase their emotional response proportionately as the number of victims increases. Slovic (2007) describes “psychic numbing” as a tendency for affective response to flatten when confronted with large numbers of victims. This creates a challenge for suffering reduction because many important problems are statistical rather than personal.

The identifiable victim effect. A single identifiable individual can evoke stronger emotional responses than a much larger but statistically described group (Small et al., 2007). This does not mean that stories are always better than statistics. Stories can also distort priorities. A responsible moral-reflection system should therefore combine concrete cases with population-level evidence rather than replacing one with the other.

Distance and abstraction. Suffering that is geographically distant, institutionally hidden, or temporally remote is often less salient than an immediately visible case. This can affect concern for refugees, animals in industrial systems, wild animals, and future beings. AI could potentially reduce some of this abstraction by translating technical evidence into accessible explanations.

Speciesism and moral inconsistency. People may use different standards when evaluating similar harms to humans and animals. Research on speciesism suggests that perceived similarity to humans, emotional attachment, and beliefs about intelligence can influence moral standing judgments (Caviola et al., 2019). This is relevant to AI-assisted reflection because a system could invite users to articulate the principle behind their judgment and then test whether the same principle is applied consistently.

These mechanisms should not be treated as proof that people are “biased” whenever they disagree with SFE. Some differences in moral judgment may reflect genuine normative disagreement rather than cognitive error. An ethical AI system should therefore distinguish between empirical biases and legitimate value differences. Its role should be to make reasoning clearer, not to manufacture agreement.

4. How LLMs Could Support Moral Reflection

LLMs have four characteristics that could make them useful in this domain: interactivity, personalization, linguistic flexibility, and scalability. These characteristics do not make an LLM morally authoritative. They simply create opportunities for structured reflection.

4.1 Socratic questioning.

Instead of telling a user what they must believe, an AI assistant can ask questions that reveal the user's own principles. For example: “What makes you think a being deserves protection from severe suffering?” followed by “Would that reason apply to a pig as well as to a dog?” This approach can reduce the confrontational character of moral debate and make disagreement explicit. The value of such dialogue should be tested rather than assumed. Recent experimental work suggests that LLMs can produce persuasive arguments tailored to individual users, which is both an opportunity and a safety concern (Salvi et al., 2024).

4.2 Evidence-grounded perspective-taking.

An LLM can translate scientific information into concrete descriptions. For example, it can explain what confinement, injury, heat stress, or separation may mean for an animal, while clearly distinguishing established evidence from inference. The goal is not to pretend to know exactly what a non-human animal “feels,” but to help users understand the relevant evidence about welfare.

4.3 Reframing large-scale suffering.

An AI system can place numbers in context: explain what a statistic measures, distinguish individuals from events, and compare the severity and duration of harms. It can also present uncertainty ranges rather than giving false precision. This may help counter the tendency for very large numbers to become psychologically meaningless.

4.4 From reflection to action.

Moral concern has little practical value if it never affects behavior. An assistant could help users identify low-cost actions consistent with their own values—for example, learning about higher-welfare food choices, supporting credible animal-welfare interventions, helping an injured animal through appropriate local services, or learning more about neglected human suffering. Recommendations should be framed as options rather than moral tests.

4.5 Educational and institutional use.

The same architecture could be adapted for classrooms, reading groups, ethics courses, or workplace discussions. The system could present competing ethical arguments, ask participants to identify assumptions, and summarize disagreements. This may be more appropriate than a “compassion coach” framing in settings where neutrality and intellectual diversity are important.

5. Conceptual Framework: CompassionGPT

This paper proposes CompassionGPT as a conceptual design rather than a finished product. The system would consist of five modules.

1. Moral Reflection Module. Generates open-ended questions based on the user's stated values. It should avoid coercive language and explicitly allow disagreement.

2. Evidence and Source Module. Retrieves information from curated sources, preferably primary research, reputable scientific organizations, and established philosophical works. Claims should be traceable to sources, and uncertainty should be displayed when evidence is incomplete.

3. Perspective Module. Produces short, scientifically grounded descriptions of relevant experiences or welfare conditions. It should avoid fabricated first-person claims such as “I am suffering” when the system is describing an animal or an uncertain case.

4. Trade-off Module. Helps users compare harms and benefits without silently assuming a single moral theory. Where the question depends on a specific ethical framework, the system should state that framework.

5. Action Module. Converts reflection into practical options and identifies potential unintended consequences. For interventions affecting animals or ecosystems, the system should recommend professional or authoritative guidance when appropriate.

A possible interaction is:

User: “I care about animals, but I have never thought much about farmed animals.”

System: “That is a common starting point. What qualities make you think a companion animal deserves protection from severe pain—its capacity to feel pain, its emotional relationships, its intelligence, or something else?”

User: “Mostly its ability to suffer.”

System: “If capacity for suffering is the main reason, would you want to apply the same principle to farmed animals? If yes, we can look at evidence about their welfare. If no, we can explore what additional difference you think matters.”

The example illustrates the intended design principle: the AI does not announce a conclusion first. It helps the user identify a principle, apply it consistently, inspect relevant evidence, and decide what follows.

The architecture should also include a “steelman” function. Before challenging a user's view, the system should present the strongest reasonable objection to its own recommendation. This reduces the risk that the tool becomes a one-sided persuasion engine.

6. Research and Evaluation Agenda

The strongest version of this proposal is empirical. A conceptual argument that an LLM could expand moral concern is not evidence that it actually does so.

A first-stage study could compare three conditions: static educational material, a neutral conversational AI that answers questions without a targeted moral objective, and a compassion-oriented AI using the framework proposed here. Participants could complete validated measures of moral expansiveness and speciesism before and after the intervention (Crimston et al., 2016; Caviola et al., 2019).

A stronger design would include behavioral outcomes. Participants could be offered opportunities to allocate a small amount of money, time, or attention among different causes. The purpose would not be to force a preferred allocation, but to test whether changes in stated concern correspond to changes in decisions.

A follow-up study should measure persistence. An immediate post-conversation increase in empathy may disappear within days. Useful endpoints could include one-week and one-month follow-ups, repeated behavioral choices, and willingness to seek additional information.

Several experimental comparisons would be especially informative:
• statistics versus narrative;
• narrative versus Socratic dialogue;
• neutral dialogue versus suffering-focused dialogue;
• one-time interaction versus repeated interaction;

• human-written versus AI-generated explanations;
• high-emotion versus low-emotion framing.

Researchers should pre-register hypotheses and distinguish exploratory from confirmatory analyses. If participants are recruited for research, informed consent and appropriate ethics review should be obtained. No claims about effectiveness should be made from hypothetical user interactions alone.

An important outcome measure is not simply “did the participant agree?” but “did the participant become better informed, more consistent, and more willing to take proportionate action?” A successful intervention should ideally increase concern without increasing gullibility, polarization, or dependence on the AI.

7. Risks, Limitations, and Ethical Challenges

The same capabilities that make LLMs useful for moral reflection make them potentially dangerous.

Persuasive overreach. A model can generate fluent arguments tailored to a particular person. If the objective is defined as “maximize compassion,” the system may learn to pressure users rather than respect their autonomy. This is unacceptable for an ethical educational tool. The objective should instead be informed reflection: present evidence, clarify assumptions, acknowledge uncertainty, and let users decide.

Factual hallucination. A model may invent claims about animal cognition, welfare conditions, nutrition, or ecological interventions. This risk is particularly serious when emotionally vivid claims are used to motivate behavior. Retrieval-augmented generation, source citations, uncertainty labels, and human review should be standard features.

Anthropomorphism. Perspective-taking can become misleading when the system writes as if it were a particular animal. First-person narratives may create emotional engagement, but they can also falsely imply knowledge of an animal's subjective experience. The safer alternative is evidence-based narrative with explicit separation between observed behavior, scientific inference, and imaginative illustration.

Bias amplification. Training data contain existing cultural, political, and species-related biases. An AI designed to expand moral concern may therefore reproduce the very exclusions it is intended to examine. Evaluation should include diverse populations, languages, ethical traditions, and views that challenge the system's assumptions.

Moral monoculture. SFE is a family of positions, not a single doctrine, and many people reject its strongest claims. The tool should therefore make its normative assumptions visible. It should be capable of presenting consequentialist, deontological, virtue-ethical, religious, and common-sense objections rather than treating one framework as settled fact.

Attitude-behavior gap. People may express greater concern without changing what they actually do. The intervention should therefore be evaluated using behavioral outcomes and longer follow-ups.

Misplaced moral attention. Users may become highly concerned about speculative future digital minds while neglecting clear and present suffering. Conversely, concern for animals could be used to dismiss serious human suffering. The system should resist zero-sum framing and help users reason about both neglected and well-recognized forms of suffering.

Future AI suffering. The possibility of artificial sentience is uncertain, and current LLMs should not be casually described as conscious sufferers. Nevertheless, the development of systems with potentially relevant forms of artificial phenomenology raises precautionary questions. Metzinger (2021) argues for a moratorium on research aimed at synthetic phenomenology, while others contest aspects of that argument. The relevant lesson here is narrower: an AI-for-compassion agenda should not ignore the possibility that future technology could create new forms of suffering.

Selection-effects. Future studies should also account for selection effects, since individuals who voluntarily engage with a compassion-oriented AI may already differ from the broader population in moral concern or openness to moral reflection.

LLM variability. Because LLM outputs can vary across models, versions, prompts, and deployment settings, any empirical evaluation should specify the model, system instructions, retrieval sources, temperature or sampling settings where relevant, and evaluation period sufficiently for replication.

AI is not a substitute for institutions. Many major sources of suffering are embedded in economic, political, legal, and technological systems. A chatbot can support reflection, but it cannot by itself reform food systems, improve labor conditions, strengthen animal protection, or govern advanced AI responsibly.

8. Discussion

The proposal has three possible interpretations. The strongest is that LLMs could actively increase compassion and thereby reduce suffering. The moderate interpretation is that they can improve people's ability to reason about neglected suffering. The weakest, but most defensible at present, is that they are useful interfaces for delivering and discussing existing evidence about moral concern.

The moderate interpretation is the most promising research target. It does not require assuming that AI possesses empathy. It treats the model as an instrument that can scaffold human reasoning in much the same way that calculators scaffold numerical reasoning or maps scaffold spatial reasoning.

This framing also changes how success should be defined. A system should not be rewarded simply for making users feel sad. Emotional intensity is not the same as moral improvement. A user might feel strong distress after reading a graphic story and still take no useful action. Worse, repeated exposure could lead to avoidance or compassion fatigue. A better system would aim for calibrated concern: enough attention to recognize serious suffering, enough reasoning to distinguish high-impact from low-impact actions, and enough practical guidance to support proportionate responses.

The proposal also suggests a broader connection between moral circle expansion and AI safety. If future AI systems become powerful social or institutional actors, their treatment of sentient beings could depend partly on the values embedded in their training, deployment, and governance. Work on moral concern today may therefore have consequences beyond present-day human and animal welfare. At the same time, attempting to encode a single moral doctrine into future systems could create its own risks. The safer research direction is to improve moral reasoning, uncertainty awareness, and sensitivity to suffering while preserving room for legitimate ethical disagreement.

The central empirical question remains open: can carefully designed conversational systems produce durable improvements in moral concern and behavior without increasing manipulation or misinformation? That is a tractable research program. It can be investigated with randomized studies, behavioral measures, longitudinal follow-up, and transparent reporting.

9. Conclusion

Large language models cannot replace human compassion, moral agency, or institutions. They can, however, provide a new interface for moral reflection. Their ability to converse, personalize explanations, translate technical evidence, and explore alternative perspectives creates a plausible opportunity to make neglected suffering more visible.

From a suffering-focused perspective, the most promising use is not to make people feel guilty or to maximize emotional reactions. It is to help people notice morally relevant suffering, examine whether their principles are applied consistently, understand uncertainty, and identify practical ways of helping.

CompassionGPT is therefore best understood as a research hypothesis rather than a finished solution. Its value depends on evidence: whether it improves knowledge, broadens moral concern, changes behavior, and does so without coercion or serious unintended harms.

The next step is not to build a system that assumes it already knows the correct moral answer. It is to build and test systems that help people reason more carefully about suffering—especially where the victims are distant, numerous, unfamiliar, or easily overlooked.

Author Contributions and Declarations

Author contributions. Suvarna Ahire led the conceptualization, literature synthesis, system design, and manuscript preparation. Leena Ahire and Shailendra Ahire contributed to development and review of the conceptual framework and manuscript. All authors approved the final manuscript.

Funding. No external funding is reported for this work.

Conflict of interest. The authors declare no known financial or personal conflicts of interest relevant to this manuscript.

Human participants and data availability. This is a conceptual research paper and does not report an original human-participant study, survey, or proprietary dataset. No human-subject data were collected for the analyses presented here.

Use of generative AI. Generative AI tools were used during drafting and editing assistance. The authors are responsible for the final text, interpretation of the literature, citations, and claims presented in the manuscript.

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